

An effective obstacle detection system using deep learning advantages to aid blind and visually impaired navigation

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ABSTRACT

Blind and visually impaired people face different challenges when navigating indoors and outdoors. In this context, we suggest developing an obstacle detection system based on a modified YOLO v5 neural network architecture. The suggested system is capable of recognizing and locating a set of landmark indoor and outdoor objects that are extremely useful for Blind and Visually Impaired (BVI) navigation aids. Training and evaluation experiments were conducted using two datasets: the IODR dataset for indoor object detection and the MS COCO dataset for outdoor object detection. We used several optimization strategies, such as model width scaling, quantization, and channel pruning, to guarantee that the suggested work is implemented in embedded devices in a lightweight manner. The proposed system was successful in achieving results that were extremely competitive in terms of processing time as well as the precision of obstacle detection.

1. Introduction

In modern life, information and assistive technologies for autonomous navigation of the Blind and Visually Impaired (BVI) present one of the most significant problems that need to be addressed. Various organizations and governments proposed new infrastructures for BVI.

The World Health Organization reports that 2.2 billion people have visual impairments or are blind, with 40 million of those individuals being completely blind [1].

One of the most usual problems that can be faced by BVI people during routine life is environmental awareness. Various options, such as cane guides, dogs, and smartphone apps, have been presented to this goal, but each has its own set of drawbacks. For example, the cane is not effective for long distances and overcrowded places and trained dogs are very expensive and require a high level of attention. Also, it is not suitable to prevent BVI from dangerous situations.

Autonomous navigation presents an extremely important factor for BVI persons. It is critical to develop innovative assistive technology to enable these people to carry out their regular duties more safely. Within

this context, building new obstacle detection systems based on Deep Learning architectures that can effectively detect indoor and outdoor obstacles is extremely desirable.

Many solutions based on Global Positioning Systems (GPS) have been presented recently. However, because of their limited accuracy and precision, as well as their inability to be used in an interior setting, these solutions are not suited for BVI residents. Furthermore, GPS cannot offer information about nearby objects. Many other solutions are proposed based on computer vision approaches to address autonomous assistance navigation for BVI persons. Simultaneous localization and mapping (SLAM) [2] techniques are proposed to contribute to new environment mapping and assist those who are blind or visually impaired with navigation and determining their location.

Building new systems for obstacle detection and avoidance can highly improve the life quality and then safety level for BVI persons. BVI persons still face various challenges during their everyday activities. For normal people, precisely detecting objects and localizing them present an extremely easy task. However, it is very challenging for the visually impaired and even impossible for blind people. To this end, it is

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increasingly valuable to design efficient obstacle detection systems to help BVI persons more integrate into society and explore more and autonomously away from their surrounding environments.

Recently, Deep Learning-based architecture has emerged to present a new set of effective solutions to solve different computer vision and Artificial Intelligence issues. It has been applied to different fields of research including pedestrian detection [3], fatigue detection [4], emotion recognition, indoor object identification [5], indoor scene recognition [6], road sign detection [7], and medical applications [8–10].

This work suggests a lightweight obstacle detection system able to detect landmark objects in indoor and outdoor environments by using deep learning approaches. The suggested system gives BVI persons the ability to move freely in both indoor and outdoor locations more safely. We note that the developed system has been trained and evaluated using two datasets: IODR dataset for indoor object detection and MS COCO dataset for outdoor object detection. The proposed work is robust as it was trained using images captured under a variety of difficult conditions, such as different lighting conditions (day and night), variations within and between classes, and objects being partially occluded or viewed from different angles.

In the proposed approach, the problem to be addressed remains is still very hard, particularly indoor, where decorations and backgrounds are often very complicated and cluttered.

The proposed obstacle detection system is developed by applying the advantages of Deep Learning techniques. The suggested technique for detecting obstacles is developed using a “YOLO v5” [11]. To ensure that the proposed system can be implemented on embedded devices efficiently, different compression and optimization techniques have been used, including model width scaling, channel pruning, and model quantization. The proposed obstacle detection system has demonstrated very competitive detection accuracies as well as a good time process. We used two model compression strategies to contribute to a lightweight implementation of the proposed work: channel pruning and post-training quantization. After optimization techniques application, we succeeded to obtain a lighter version of the developed system that can fit the memory of embedded devices.

The remainder of the paper is organized as follows: Section 2 will contain a review of related works. Section 3 will describe the proposed obstacle avoidance architecture and compression techniques used for a lightweight implementation on FPGAs. The results of experiments on two benchmark datasets (IODR and MS COCO) will be presented in Section 4. Section 5 provides conclusions.

2. Related work

Different assistive technologies and technical interventions can be developed to make the BVI lives easier and to make them more independent to navigate in new unfamiliar surrounding environments. It is incredibly exciting to create new assistive technology to assist individuals with visual impairments in fully participating in daily life activities.

Indoor and outdoor assistance navigation for BVI persons still presents very challenging artificial intelligence and computer vision fields. It is critical to develop assistive technologies which significantly enhance the life quality and independence of BVI persons.

Vision loss and blindness present one of the most frequent health problems around the world. About 80% of the persons that suffer from low vision and blindness are from middle to low-income countries which makes them not able to use assistive costly devices. Apart from the medical treatment, this category of persons can use different assistive technologies such as white canes and guide dogs. Also, they can use GPS-equipped devices in outdoor environments but this technique is not suitable for the indoor environment. In [12] authors proposed to develop an Electronic Mobility Cane (EMC) used to help visually impaired people to detect and avoid obstacles. The developed system

constructs a logical map to provide the visually impaired user with information about the surrounding environment. A haptic cane is proposed in [13], it is equipped with different smart sensors to provide the user with obstacle information. An Electronic Travel Aids (ETA) system is proposed in [14], it is equipped with a multi-sensor and a pair of eyeglasses that help the impaired person to navigate in a safer way using ultrasonic and depth sensors. In [15], authors proposed to design a smart cane used to detect obstacles for visually impaired persons. The authors have integrated a Linear-Quadratic Regulator (LQR) based controller to optimize position tracking. Takatori et al. [16] proposed a white cane based on IC tags. It supports independent walking of visually impaired users in indoor spaces. It ensures vibration feedback with a voice prompt. In [17], an intelligent system is used to help visually impaired persons to navigate in indoor environments based on a radio frequency identification (RFID) system. An object detection system is proposed in [18]. It is designed using a deformable grid (DG) and can detect objects and detect the risk of collision.

In recent years, architectures based on Deep Learning have achieved remarkable success in addressing various challenges in the computer vision field. Different assistive technologies have been developed to assist BVI in their navigation by avoiding obstacles. In [19], a smartphone-based application was proposed for detecting obstacles in both indoor and outdoor spaces. A new electronic device used for automatic navigation assistance by detecting different types of obstacles is proposed in [20]. The designed system is named “NavCane” which is equipped with different types of sensors that contribute to detecting obstacles. In [21], the authors suggested creating a new assistive technology employed to help visually impaired people. The system can perform a semantic characterization of the classification of obstacles. In [22,23] authors use the advantages of Deep Convolutional Neural Networks (DCNNs) to build an effective assistance technique. These systems can identify a range of important indoor objects that can assist BVI individuals with navigation.

3. Proposed method

The visual sense presents one of the most important senses for a human being. Our visual system is extremely complex and powerful as it enables us to detect, recognize, and perceive the objects surrounding us. Although the human visual system is complex, it is still challenging to develop new assistive systems based on object detection and obstacle avoidance used to help BVI to facilitate for them more their life and to contribute to safer navigation for BVI.

To help BVI people navigate in unfamiliar indoor and outdoor areas, in the following, we suggest developing a deep learning-based obstacle detection system. During the development of the suggested system, we tacked into account the following facts:

- Training and testing the developed system using unsafe situations (upstairs, downstairs, heavy occlusion, crowded images, etc...).
- High valuable object classes for BVI assistance navigation.
- Training and evaluation processes have been conducted under very challenging conditions.
- The proposed system has been trained and tested using two datasets: IODR [24] and MS COCO [25].
- The obstacle detection system can detect a set of new landmark objects that were not considered in the state-of-the-art works.

Deep learning architectures have advanced significantly in recent years, making them a primary solution for unsolvable tasks. Visual impairments and blindness can lead to social isolation. To solve this problem, we suggest building an obstacle detection system to help BVI people integrate into daily life and make life easier. The system being suggested is built using the one-stage neural network “YOLO v5”.

YOLO v5 architecture is very similar to YOLO v4 architecture but with some minor modifications. YOLO v5 architecture presents new

modifications in the architecture that contributed to enhancing the speed and detection accuracy while reducing the network computation complexity. This architecture uses a new backbone that changes the features process as well as optimizations techniques. This architecture proposes a set of new efficient techniques that enhance the training process. Also, it modified the detection head which enhanced the accuracy. This architecture applies different data augmentation techniques to enhance the generalization of the model for different applications and tasks. YOLO v5 applies the use of photometric distortion techniques to adjust brightness, contrast noise, saturation, and so on. The geometric distortion technique was applied by using flipping, rotation, scaling, and image cropping. It also applies many other data augmentation techniques as mix-up techniques and cut_mix techniques. All the used data augmentation techniques are used to increase the model's robustness against data corruption.

Early CNN models used the dropout technique to prevent the network from overfitting. This technique is applied to fully connected layers as it presents the most network portion that presents neurons and requires huge computation resources. However, this technique cannot be applied to the convolution layers. For this, in YOLO v5 the drop block technique is proposed. So, instead of dropping single neurons, a block of neurons with a fixed size is dropped. The drop block method was introduced to combat the major drawback of dropout being dropping features randomly which proved to be an effective strategy for fully connected networks but less fruitful when it comes to convolutional layers wherein features are spatially correlated. Fig. 1 provides the drop block technique process compared to the dropout technique.

Another strength point in the YOLO v5 architecture is the activation function. It applies the use of a sigmoid weighted linear unit (SILU). This function is calculated as follows:

$$f(x) = x\sigma(x) \quad (1)$$

A new backbone is proposed in the YOLO v5 architecture to reduce the training problems, reduce the model complexity, and ensure that the designed model can be run efficiently on devices with limited resources. The cross-stage partial network CSPNet [26] was deployed to extract valuable features from the input images. This backbone has shown a significant improvement in time processing.

The YOLO v5 is based on three main parts:

- Network backbone.
- Network neck.
- Network head.

Model backbone: the CSPNet-based backbone was adopted. Its main goal is to reduce the amount of computation by enabling the network architecture to perform a gradient combination. This purpose is achieved by dividing the input features map into two parts. After this, it merges them through the cross-stage hierarchy. CSPNet enables the gradient flow to propagate through different paths. In this way, the

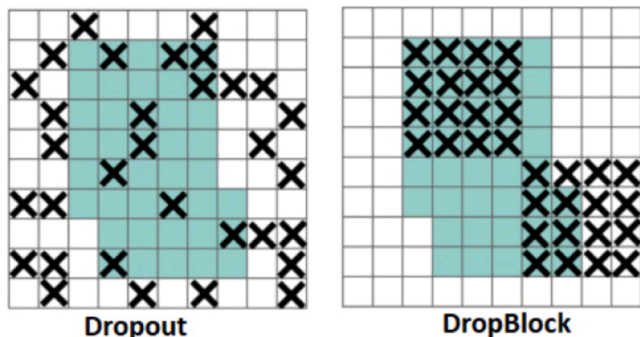


Fig. 1. Difference between dropout and drop block techniques.

gradient information presents a large correlation difference by switching from transition to concatenation. The CSPNet can widely reduce the amount of computation and improve the network accuracy as well as the inference speed. By applying the CSPNet, three main contributions can be achieved:

- Increasing the learning ability of the CNN model.
- Reduce the computation complexity.
- Reduce memory costs.

Model neck: the model neck is used to generate the features pyramid which helps the model to detect objects at different scales. The model neck enables the network to identify the same objects with different sizes and scales. YOLO v5 uses the Path Aggregation Network (PAN) [27] to generate the features pyramid. This network introduced a new shortcut connection that enables the low-layer features to propagate easier to the top layers. The PAN network is based on the Features Pyramid Network (FPN) with a modified shortcut connection. In the FPN network, the spatial information is passed through the (red path) while in the PAN, spatial information passes through less number of layers to reach the top layers (green path). To collect feature information, PAN adds all the previous layers to the current layers. Fig. 2 provides the PAN network architecture.

Another strength of the YOLO v5 network is the focus module which is used to slice the input image into four small images and concatenate them to apply the convolution operation. YOLO v5 architecture adopts an automatic algorithm used for learning bounding box anchors which calculates the anchor boxes' size at different image sizes and scales and highly contributes to the detection accuracy. The Generalized Intersection over Union (GIOU) is adopted to calculate the regression loss as presented in the following Eq. (2).

$$L_{GIOU} = 1 - IOU + \frac{|CB \cup |}{|C|} \quad (2)$$

B^{gt} is the ground truth box.

B is the predicted box

$$IOU = \frac{B \cap B^{gt}}{B \cup B^{gt}}$$

C is the smallest box that covers B and B^{gt} .

GIOU provides a more comprehensive evaluation of bounding box predictions by considering both overlap and extent, leading to improved object localization, shape robustness, and evaluation consistency. These advantages make it a popular choice in the computer vision community for various tasks involving object detection and localization.

Network head: for the detection process, YOLO v5 performs a 1×1 convolution at three scales to detect objects at three scales (small, medium, and large). Fig. 3 provides the detailed architecture of YOLO v5 adopted in the proposed work.

To ensure a lighter version of YOLO v5 that can easily fit an embedded device memory, we propose to change the number of output channels of the model which can widely impact the model calculation complexity as well as the model memory storage. For this purpose, we applied a model width scaling. YOLO v5 presents different versions as YOLO v5l, YOLO v5m, YOLO v5s, and YOLO v5x. We note that YOLO v5x is the most effective version in terms of detection precision and YOLO v5s is the faster version.

In the proposed implementation, we used the same model depth as in the YOLO v5s version (0.33) and we changed the model width from 0.5 in YOLO v5 to 0.33 in the proposed work. By applying the model width scaling techniques, we ensured a big drop in the model complexity as well as high speed for the model processing time.

Other optimizations and compression techniques are used to guarantee a lightweight implementation in embedded systems. In this context, we applied pruning and quantization techniques.

channel pruning strategy [28] was applied, which tries to pick the

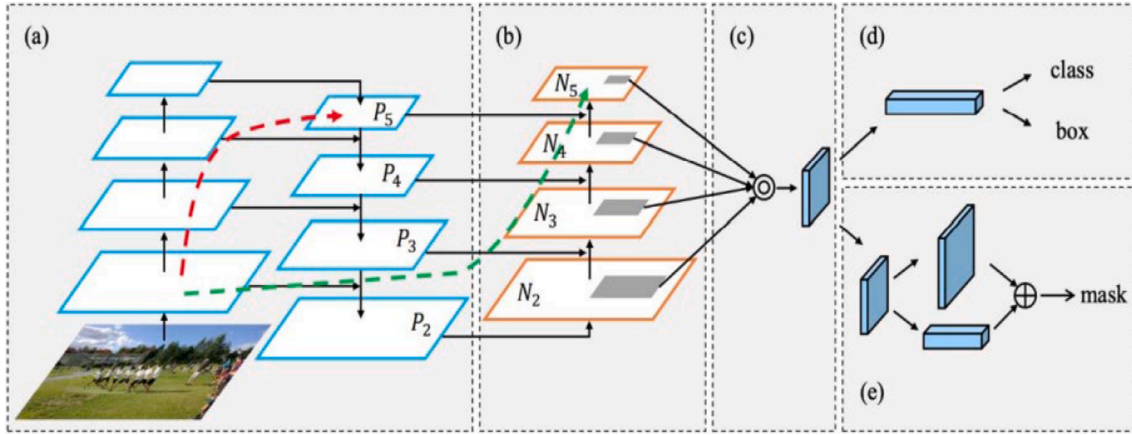


Fig. 2. PAN network architecture.

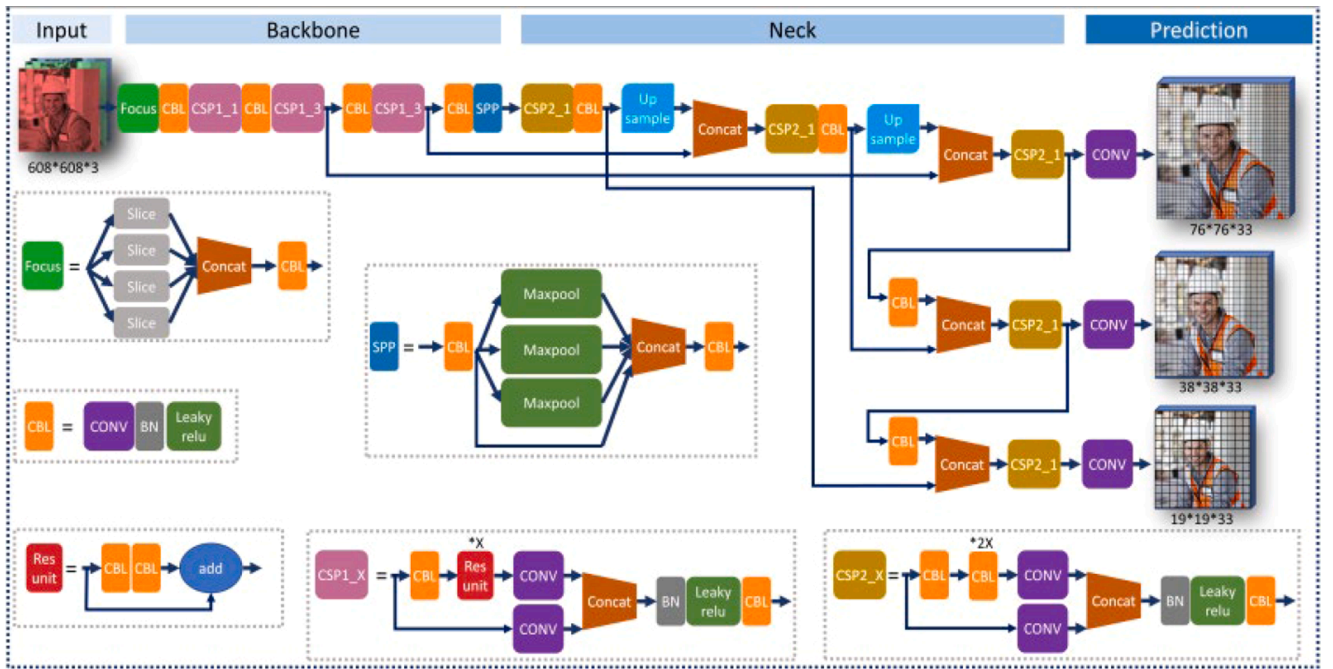


Fig. 3. YOLO v5s architecture.

important channels that contribute to model accuracy while eliminating the useless channels. The pruning technique was utilized following the completion of training. The post-training channels pruning technique has been applied based on two steps: first, the channel selection algorithm has been applied based on the regression loss. By using this technique, the relevant channels are maintained while the weak channels are removed. Secondly, the feature maps of each layer are reconstructed using the least square method. After this, a fine-tuning of the pruned model is applied to recover the model's accuracy.

The second technique applied to reduce the size of the model is the post-training quantization technique [29]. It is applied to transform the activation and the weights representation from 32 bits floating points to 2 bits fixed points. After the training process, the quantization technique was applied to the model and the straight-through estimator method [30] was used for fine-tuning to improve accuracy.

4. Results of experiments

4.1. Experiments for training and testing

Obstacle detection presents one of the most active fields of research especially when it is applied to assistance navigation for blind and sighted persons. It is extremely important to correctly detect and localize objects for this category of persons to fully help them during their navigation in new indoor and outdoor unfamiliar environments.

The proposed experiments were carried out using a desktop computer with an Intel i7 CPU, 32 GB of RAM, and an NVIDIA GTX 960 GPU with 4 GB of graphic memory. For the development of the proposed obstacle detection system, the PyTorch Deep Learning framework has been used with CUDA support and cuDNN library support. For image processing and manipulation, the open computer vision OpenCV library has been used.

To ensure a robust system that can detect the maximum of possible objects that can face the BVI person during its navigation, experiments for training and testing were performed using two datasets: Indoor Objects Detection and Recognition dataset (IODR), which was used to

Table 1

Number of parameters and FLOPs before and after compression techniques application.

Model	Parameters (M)	FLOPs (B)
Proposed YOLO v5 before compression	2.1	4.8
Proposed YOLO v5 after compression	1.7	4.2

detect 21 landmark indoor objects, and MS COCO dataset, which was used to detect a set of other indoor and outdoor objects. We note that we select from the MS COCO dataset only the valuable objects that can present obstacles to be avoided during BVI navigation.

Training and testing experiments were conducted using images that were captured under challenging conditions, including a variety of lighting conditions, occlusion, different perspectives of objects, and high variations within and between classes which can highly contribute to building a robust obstacle detection system with high detection performances. The model has been trained for epochs, each epoch with 10,000 iterations. The value has been 0.001.

We address in this paper the obstacles and dangers avoidance to reach safely and independently their desired destinations. The selected objects used to build the proposed system are essential for enhancing the quality of life for individuals with visual impairments and fully assist them to more explore their indoor and outdoor surroundings.

The proposed obstacle detection system presents one step further to ensure a high safety degree for the BVI around the world by taking advantage of Deep Learning-based architectures. During the training and testing experiments, we only choose to use some specific object classes that are highly valuable to assist BVI to detect obstacles and to ensure more safety when accessing new unfamiliar environments. The IODR dataset provides 8000 indoor images containing 16 different indoor landmark objects and the MS COCO dataset provides 91 generic objects and 328,000 images. During the proposed experiments, we only focus our attention on 40 object classes that can contribute to the BVI assistance navigation and obstacle detection. In order to ensure a lightweight implementation of the proposed work, we applied different optimization techniques including channel pruning and quantization.

We used the channel pruning technique to keep the important channels and delete the useless ones. This technique was used following the training procedure. We used the post-training quantization technique to assure a lighter version of the suggested work. This technique aims to replace the data presentation from 32 bits floating point to 2 bits fixed points representation for the activation and weights presentation.

In summary, the proposed obstacle detection system has undergone different optimization as channel pruning and quantization to ensure a compressed version of the proposed work.

By applying the different optimizations, we succeeded to reduce the number of model parameters and FLOPs. The following Table 1 provides the number of parameters and FLOPs before and after pruning and quantization.

We note that based on Table 1 the proposed system is suitable to fit the memory of FPGAs or low-end devices. The proposed compression

technique has reduced the number of parameters from 2.1 million to 1.7 million. Besides, the number of FLOPs was reduced from 4.8 to 4.2 which will accelerate the processing speed. Based on the achieved results, the proposed model presents high applicability in low-end devices with limited computation resources. The compression technique enhanced the processing speed but degraded the detection accuracy. The achieved results will be discussed in detail in the following.

Table 2 provides the obtained per-class average precision of YOLO v5 before optimization techniques are applied in the IODR + MS COCO datasets.

As provided in Table 2, we obtained very encouraging detection precision in the objects selected from IODR and MS COCO datasets for almost all the class objects. We note that these detection results are achieved only by applying the proposed optimization of the model width scaling. We achieved more than 80% as a per-class average precision for almost all the object classes and we achieved 81.02% as mean Average Precision (mAP) for all the object classes.

Table 3 reports the per-class detection precision on the IODR + MS COCO datasets after compression and optimization techniques application.

As reported in Table 3, we obtained very encouraging detection results despite the optimization techniques application. We succeeded to obtain very competitive detection accuracies for almost all the object classes that can highly contribute to increasing the safety level of indoor and outdoor assistance navigation for BVI persons by detecting different obstacles in a very precise way to save this category of persons from dangerous situations.

As presented in Table 3, we achieved very interesting per-class detection results after the channel pruning and the post-training quantization technique application. Considering the compression process, it is expected that there is an impact on the model's performance, as the compressed model may have a reduced capacity to accurately detect objects compared to the original uncompressed model. This could explain the differences observed between AP percentages presented in Table 3 and Table 4.

The reduction in model size through compression techniques often involves sacrificing some level of model complexity or information. This can lead to a trade-off between model size and performance. In this work, compression techniques introduced slight degradation in model accuracy while enhancing the processing speed and making it suitable for implementation on low-end devices.

Table 4 reports the mean average precision (mAP) obtained for IODR + MS COCO datasets before and after optimization techniques applications. We obtained 76.41 as a mean Average Precision (mAP) which refers to an accuracy drop of 4.6 compared to Table 2. We obtained very competitive detection accuracies for almost all the object classes before and after the optimization techniques application. We note that we achieved high accuracy in detection but with low accuracy loss after channel pruning and post-training quantization.

Table 4 shows that the suggested YOLO v5 and compressed versions have extremely good detection precision and processing time. We achieved 81.02% mean average precision for the proposed YOLO v5 and 67

Table 2

Per-class detection precision obtained before compression in IODR + MS COCO datasets.

Class name	Electricity box	Sign	Window	Trash can	Notice table	Door	light	Elevator
AP(%)	79.8	81.2	80.7	81.3	81.2	80.9	79.5	81.6
Class name	Heating	Fire extinguisher	Stairs	Chair	Security button	Smoke detector	Light switch	Table
AP(%)	80.7	81.1	81.4	81.9	80.9	79.7	79.9	82.2
Class name	Bus	Train	Person	Fire hydrant	Bicycle	moto cycle	truck	Car
AP (%)	81.3	81.8	81.3	79.6	80.4	79.8	81.2	81.6
Class name	Mirror	Dining table	Vase	Stop sign	Couch	Bed	Street sign	Potted plant
	80.4	80.1	80.9	80.7	81.3	82.3	81.2	81.2
Class name	TV	Refrigerator	Laptop	Bench	microwave	Desk	Parking meter	Oven
AP (%)	82.5	80.7	81.4		81.9	82.1	81.1	80.8

Table 3

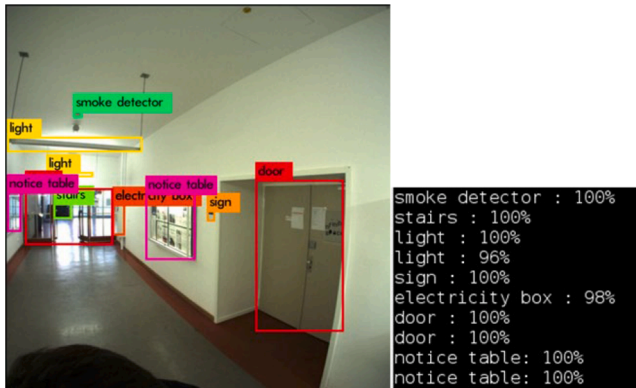
Per-class detection precision obtained after compression in IODR + MS COCO datasets.

Class name	Electricity box	Sign	Window	Trash can	Notice table	Door	light	Elevator
AP(%)	77.4	78.1	77.3	78.6	78.9	78.1	76.5	78.7
Class name	Heating	Fire extinguisher	Stairs	Chair	Security button	Smoke detector	Light switch	Table
AP(%)	77.9	77.3	79.3	79.1	76.5	77.2	77.6	79.9
Class name	Bus	Train	Person	Fire hydrant	Bicycle	moto cycle	truck	Car
AP (%)	79.2	77.5	77.9	76.9	77.6	76.4	78.4	78.8
Class name	Mirror	Dining table	Vase	Stop sign	Couch	Bed	Street sign	Potted plant
AP (%)	77.4	77.9	77.8	77.8	79.1	80.1	78.3	77.3
Class name	TV	Refrigerator	Laptop	Bench	Microwave	Desk	Parking meter	Oven
AP (%)	79.9	77.3	78.6	78.3	79.1	79.8	78.1	77.6

Table 4

Obtained mean Average Precisions and processing time.

Model	mean Average Precision (mAP)	Frame Per Second (FPS)
Proposed YOLO v5	81.02 %	67
Proposed YOLO v5 (compressed)	76.41%	89

**Fig. 4.** Detection example.

FPS as a processing time while we obtained 76.41% for the compressed version and 89 FPS as a processing time. Fig. 4 provides a detection example for indoor navigation.

4.2. Study of ablation

In this section, an ablation study is proposed to evaluate the effectiveness of the proposed work contribution. To contribute to an obstacle detection system deployable in embedded systems, we performed a width scaling method. We maintained the same model depth as in the YOLO v5s version and we manipulated the model width using different values. We used different model widths which refer to the number of the model's output channels. We performed different experiments to achieve a better model width that fits the proposed work requirements. We adopted different model width values including 0.45, 0.40, 0.35 and 0.30. We note that for every used value of the model width, we performed new training and testing of the model using the 40 object classes chosen from the two datasets: IODR and MS COCO. After excessive experiments, we noticed that the 0.30 value of the width scaling achieved the best tradeoff between detection precision, model size, model complexity, and processing time.

As mentioned from the obtained results in section 4.1, despite the challenging conditions of the training and evaluation experiments, We obtained really intriguing detection findings that will aid BVI residents in carrying out their regular activities in a safer and less reliant manner. Considering the above results, the new assistive technology is deployable in embedded devices to assist BVI persons to fully participate in daily life.

5. Conclusion

In this paper, we proposed a new assistive technology system that can help Blind and Visually Impaired (BVI) people navigate indoor and outdoor environments more safely. This system can be used to detect obstacles and help the user move around more confidently. Two datasets for training and testing are used to evaluate our obstacle detection system. In fact, the IODR dataset is used for indoor obstacle detection, and the MS COCO dataset is employed for outdoor obstacle detection. The proposed system is robust enough since it was trained and tested under extremely difficult situations to deal with real-world scenarios, particularly during BVI people's navigation in unknown locations. It was developed based on the one-stage neural network YOLO v5. Our system enables BVI persons to recognize a set of vital landmark objects highly valuable to assist them in autonomously navigating and exploring new unfamiliar environments. The results of the experiment revealed that the designed obstacle detection system produced highly encouraging results in terms of processing time and accuracy of detection.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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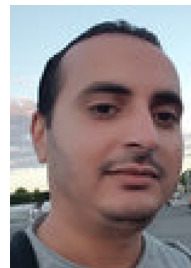
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